

# STATEMENT OF WORK (SOW)

## FOR THE MANUFACTURE AND MODIFICATIONS OF THE

NASA STANDARD INITIATOR - TYPE 1

P/N SEB26100001-2XX

NASA STANDARD INITIATOR – TYPE 3A

P/N SEH26400001-7XX

NASA STANDARD DETONATOR

P/N SEB26100094-2XX/-3XX

NASA Lyndon B. Johnson Space Center

March 2025

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## 1.0 SCOPE

### 1.1 General

This Statement of Work (SOW) establishes the requirements for the procurement and post-procurement modifications (addition of weld washers and staking) of the following devices:

- NASA Standard Initiator, Type 1 (NSI-1): P/N SEB26100001-2XX
- NASA Standard Initiator, Triple Thread (NSI-3A): P/N SEH26400001-7XX
- NASA Standard Detonator (NSD): P/N SEB26100094-2XX/-3XX

*Note: The generic use of “devices” and/or “initiators/detonators” in this SOW indicates the requirement applies to the NSI-1, NSI-3A and the NSD. The generic use of “initiators” indicates that the requirement applies only to the NSI-1 and NSI-3A.*

### 1.2 Background

The NASA Standard Initiator (NSI) was originally designed and qualified as the Single Bridgewire Apollo Stand Initiator (SBASI) to meet the requirements of the Apollo Lunar Mission. The NASA Standard Detonator (NSD) was also developed and qualified for Apollo. Both devices were subsequently adopted and standardized for use on the NASA Space Shuttle System. Currently, NSIs and NSDs are used on a variety of programs. NSI's are incorporated into the NASA Docking System (NDS) which will dock to the International Space Station (ISS), Orion uses NSI-3As, Commercial Crew Program (CCP) uses NSI-1s and NSDs, Space Launch Systems (SLS) uses NSIs, and Jet Propulsion Laboratories has NSIs on the surface of Mars, Saturn's moon Titan, and on Jupiter. Various other smaller programs through other NASA facilities, such as Wallops Flight Facility (WFF) and Goddard Space Flight Center (GSFC) also use NSIs and NSDs on unscrewed missions.

### 1.3 Device Definitions

The devices are defined as follows:

- **NSI-1:** The NSI-1 is a two-pin, hot-wire, electro-explosive device designed to convert electrical energy into a pyrotechnic output which can initiate another pyrotechnic device or train. The unit is used alone or in conjunction with other pyrotechnic elements depending on the required final output.
- **NSI-3A:** The NSI-3A configuration is identical to the NSI-1 with the exception of the mating electrical connector and connector seal.
- **NSD:** The NSD converts the deflagration input of NSI-1 or NSI-3A into a detonation output.

## **1.4 Manufacturing Sources**

All applications where the NSI and NSD are used are considered Crit I where failure to function, or inadvertent function, results in either a loss of mission and/or loss of life/crew. Mission success and crew safety are directly dependent on the critical function these devices.

Due to the criticality of the NSI and NSD, contractors must undergo a rigorous vetting process before becoming qualified to manufacture these devices for NASA. Since 1966, only a few contractors have qualified to manufacture the devices. Chemring Energetic Devices (CED) in Downers Grove, IL is currently the only qualified vendor to provide the NSIs and NSDs necessary to support NASA Missions.

## **1.5 Item Quantity**

The delivery quantity will be 200, 500 or 1000 each of the above mentioned item(s). The exact quantity will be provided by NASA in the delivery order. Lot number designators will be assigned by the NASA-JSC Contracting Officer's Representative (COR).

*Note: The Contractor will need to manufacture additional units within each lot to account for attrition and Destructive Lot Acceptance Testing (DLAT). Typical manufactured quantities range between 500-1500 devices with a maximum quantity not to exceed 1500 units per lot.*

For post-production modifications, the minimum quantity will be 100 initiators. The exact quantity will be provided by NASA in the order.

## **1.6 Milestone-Based Payments**

For milestone-based payments, the following are the proposed milestones. Information regarding the reviews is in Section 4.1.1.

- Phase I Review Complete
- Phase II Review Complete
- DLAT Successfully Completed and Report Provided
- Phase III Review Complete and Order Shipped

## **2.0 APPLICABLE DOCUMENTS**

The following documents are applicable to the manufacturing and testing of the NSI-1, NSI-3A and NSD. Drawings are available from the NASA-JSC Engineering Drawing Control Center (EDCC) at 281 483-4014.

## 2.1 NASA-JSC Device Drawings

| DOCUMENT    | REV | DOCUMENT TITLE                               | NSI-1 | NSI-3A | NSD |
|-------------|-----|----------------------------------------------|-------|--------|-----|
| SEB26100001 | L   | Initiator, NASA Standard                     | X     |        |     |
| SEB26100002 | M   | Initiator Subassembly, NASA Standard         | X     |        |     |
| SDB26100003 | F   | Initiator Body, Single Piece                 | X     |        |     |
| SLH26400001 | B   | NASA Standard Initiator, Triple Thread       |       | X      |     |
| SLH26400002 | A   | Initiator Sub-Assembly, Triple Thread        |       | X      |     |
| SDH26400003 | NC  | Initiator Body, Triple Thread                |       | X      |     |
| SEB26100005 | G   | Pin Header Assembly – Glass Seal             | X     | X      |     |
| SDB26100006 | E   | Header Body, Two Pin Initiator               | X     | X      |     |
| SDB26100007 | B   | Pin, Glass Seal, Header                      | X     | X      |     |
| SDB26100008 | C   | Disk, Sealing                                | X     | X      |     |
| SDB26100009 | F   | Charge Cup, NASA Standard Initiator          | X     | X      |     |
| SDB26100010 | B   | Disk, Insulating                             | X     | X      |     |
| SDB26100011 | B   | Washer, Insulating                           | X     | X      |     |
| SDB26100013 | C   | Cup, Closure                                 | X     | X      |     |
| SDB26100014 | A   | Washer, Sealing                              | X     | X      |     |
| SDB26100016 | C   | Marking Drawing, Initiator                   | X     | X      |     |
| SDB26100022 | A   | Specification for Propellant, NSI            | X     | X      |     |
| SDB26100023 | A   | Specification for Bridgewire, NSI            | X     | X      |     |
| SDB26100024 | B   | Spec. for Adhesive, Charge Cup/Body, NSI     | X     | X      |     |
| SDB26100025 | B   | Spec. for Adhesive, Charge Cup/Pin Seal, NSI | X     | X      |     |
| SDB26100026 | B   | Spec. for Potting, Electrical Connector, NSI | X     | X      |     |
| SDB26100077 | D   | Initiator, Body 3/8-24 Thread                | X     |        |     |
| SEB26100078 | B   | Body Assembly, Welded                        | X     | X      |     |
| SDB26100079 | C   | Shell, Connector                             | X     |        |     |
| SDD26100095 | A   | O-Ring, NSI Interfaces                       | X     | X      |     |
| SEB26100094 | M   | Assembly – NSD                               |       |        | X   |
| SDB26100089 | C   | Disk – NSD                                   |       |        | X   |
| SDB26100091 | B   | Spacer – NSD                                 |       |        | X   |
| SDB26100092 | E   | Plug – NSD                                   |       |        | X   |
| SDB26100093 | C   | Body – NSD                                   |       |        | X   |
| SLB26100052 | K   | Configuration Control Drawing, NSI           | X     |        |     |
| SLH26400052 | A   | Staking Configuration, NSI, Triple Thread    |       | X      |     |

## 2.2 NASA-JSC Informational Drawings

| DOCUMENT    | REV | DOCUMENT TITLE                               | NSI-1 | NSI-3A | NSD |
|-------------|-----|----------------------------------------------|-------|--------|-----|
| SEB26100060 | L   | Faraday Cap Assembly                         | X     |        |     |
| SLH26400060 | NC  | NSI-3A Shorting Plug                         |       | X      |     |
| SAD26100083 | A   | Drawing Tree, NASA Standard Initiator        | X     | X      |     |
| SLB26100054 | C   | Wrench, NSI, Installation Tool               | X     |        |     |
| SED26100128 | B   | Initiator Firing Unit (IFU)                  | X     | X      | X   |
| SLD26148200 | NC  | Initiator Resistance Measurement Unit (IRMU) | X     | X      | X   |
| SEB26100021 | H   | Fixture, 10cc Bomb                           | X     |        |     |
| SED26100121 | C   | Test Vessel, NSI, 1 cc Closed Bomb           | X     |        |     |
| SDD38112279 | C   | Sleeve, Detonator (Dent Test)                |       |        | X   |
| SDD38112280 | C   | Block, Detonator (Dent Test)                 |       |        | X   |

## 2.3 NASA-JSC Design and Performance Specifications

| DOCUMENT    | REV | DOCUMENT TITLE                           | NSI-1 | NSI-3A | NSD |
|-------------|-----|------------------------------------------|-------|--------|-----|
| SKB26100066 | H   | Design and Performance Spec. for NSI-1   | X     | X      |     |
| SKB26100097 | J   | Design and Performance Spec. for the NSD |       |        | X   |

## 2.4 NASA-JSC Other Applicable Documents

| DOCUMENT    | REV | DOCUMENT TITLE                            | NSI-1 | NSI-3A | NSD |
|-------------|-----|-------------------------------------------|-------|--------|-----|
| JSC 20431   | A   | JSC Neutron Radiography Specification     | X     | X      | X   |
| JSC 28035   | B   | PRACA Requirements for JSC GFE            | X     | X      | X   |
| JSC 28596   | A   | NASA Standard Initiator User's Guide      | X     |        |     |
| JSC 62809   | G   | Human Rated Spacecraft Pyro Specification | X     | X      | X   |
| SKD26100087 | A   | NSI Manufacturer Qualification Plan       | X     | X      |     |

## 2.5 Other Applicable Documents

| DOCUMENT              | REVISION | DOCUMENT TITLE                                                                  |
|-----------------------|----------|---------------------------------------------------------------------------------|
| ISO/IEC 17025:2017(E) | 2017-11  | General Requirements for the Competence of Testing and Calibration Laboratories |
| SAE AS9100            | D        | Quality Management Systems                                                      |

## 3.0 REQUIREMENTS

### 3.1 Design Requirements

#### 3.1.1 Production Design Requirements

The Contractor shall manufacture the devices in accordance with the respective NASA Drawings listed in Section 2.1 of this SOW. The Contractor shall comply with the referenced drawings with the same force and effect as if they were given in full text.

#### 3.1.2 Post-Production Modifications Design Requirements

##### 3.1.2.1 Weld Washer

The Contractor shall install a weld washer in the initiator devices in accordance with the respective NASA Drawings listed in Section 2.1 of this SOW. The Contractor shall comply with the referenced drawings with the same force and effect as if they were given in full text.

##### 3.1.2.2 Staking

The Contractor shall stake the devices in accordance with the respective NASA Drawings listed in Section 2.1 of this SOW. The Contractor shall comply with the referenced drawings with the same force and effect as if they were given in full text.

### **3.1.2.3 Remarking**

The Contractor shall remark the initiator device after completion of 3.1.2.1 and/or 3.1.2.2 in accordance with the respective NASA Drawings listed in Section 2.1 of this SOW.

## **3.2 Performance & Acceptance Requirements**

The Contractor shall manufacture the devices in accordance with the respective Design and Performance Specification listed in Section 2.3 of this SOW.

## **3.3 Manufacturing Requirements**

### **3.3.1 Lot Manufacturing**

The Contractor shall manufacture the lot in continuous manufacturing process. All piece parts within a production lot shall be manufactured or procured from a single lot.

*Note: A Lot is a group of pyrotechnic devices (NSI or NSD) containing components which have the same design, construction, and material, fabricated (controlled by one unchanged set of baseline documents) in an essentially continuous manufacturing process (i.e., no manufacturing interruption exceeding thirty (30) days). A change in qualified personnel and equivalent equipment is allowed during manufacture of a lot.*

### **3.3.2 Gas for Cleaning**

The Contractor shall not use shop air, defined as compressed air not filtered and dried, for cleaning purposes.

### **3.3.3 Shop Aids**

The Contractor shall not use debris generating shop aids during manufacturing.

### **3.3.4 Liquids Contamination Prevention**

Except for the liquid used to make the slurry, the Contractor shall control liquids so they do not come in contact with the explosive mix. To prevent contamination of the explosive material(s), the Contractor's applicable manufacturing procedures shall specify and verify the following:

- The immediate area of explosive loading shall be free of methanol, Freon, solvents and alcohol.
- Prior to loading any explosive material, each component shall be completely dry.
- After the loading any explosive material and prior to sealing, liquids shall not to be utilized for cleaning or sealing preparation.
- After sealing and prior to leak testing, if liquids are used to clean a loaded unit, any unit failing leak testing shall not be reworked, and the unit shall be rejected.



- In the event that spillage of explosive material necessitates cleaning of the loading area with liquids, all parts must be removed from the area until after such cleaning is complete, the area is completely dry, and the liquids have been removed.

### **3.4 Propellant Requirements**

#### **3.4.1 Unloaded Samples**

The Contractor shall provide the following unloaded samples to JSC prior to the delivery order closure of last lot of initiators for the propellant blend:

- A one (1) gram sample of the Zirconium (wet) for each propellant blend.
- A one (1) gram sample of the Potassium Perchlorate for each propellant blend.
- A one (1) gram sample of the blended propellant. This sample shall be taken just prior to the weighing of the propellant for loading.

#### **3.4.2 Unloaded Sample Identification**

The Contractor shall identify all samples by name, lot number or other identifying number, and date the sample is taken. The lot number or identifying number shall be shown on the container from which the sample is taken.

#### **3.4.3 Unloaded Sample Shipment**

The Contractor shall hold the unloaded samples in Contractor bonded storage until the manufacture of the last lot of initiators has been completed. The Contractor shall then ship all of the samples to NASA-JSC.

### **3.5 Government Supplied Equipment (GSE) and Components (GSC)**

NASA-JSC shall provide GSE and GSC in accordance with the respective Design and Performance Specification listed in Section 2.3 of this SOW.

### **3.6 Lot Acceptance/Certification Requirements**

The Contractor shall submit the lot for acceptance to NASA-JSC after completion of all manufacturing, acceptance testing, and data reviews. The Contractor shall package the devices for shipment after completion of the final phase review. NASA-JSC shall issue a flight lot certification for each accepted lot of NSIs and/or NSDs.

## **4.0 REPORTING REQUIREMENTS**

### **4.1 Manufacturing Methods and Testing**

The Contractor shall delineate the manufacturing and testing requirements in the appropriate Contractor documents (e.g. drawings, procedures, specifications). These documents shall be approved by the NASA-JSC COR and Safety & Mission Assurance (S&MA) representative prior to, or during, the phase review establishing the control documentation. Any changes to these documents after the phase review shall have COR

and QE approval prior to use of the new revision. The Contractor shall provide these documents per Section 5.1 (DRD 1).

#### **4.2 Design and Fabrication of Tooling**

The Contractor shall furnish tooling for pyrotechnic loading of the devices. The Contractor shall submit design drawings to the COR prior to the first loading operation per Section 5.1 (DRD 1). Tooling used in fabrication of the initiators and detonators shall be controlled by part number, revision and serial number (S/N), if applicable, to clearly show traceability of tooling used. Where tooling is called out in a manufacturing procedure or acceptance test procedure, the serial number (S/N), if applicable, used shall be recorded in the traveler/procedure to maintain traceability.

#### **4.3 Key Personnel**

The Contractor shall provide the NASA-JSC a certification of key personnel responsible for management of the engineering, manufacturing, and quality functions required for accomplishing the requirements of this contract. If any personnel changes are made during the contract the Contractor shall notify the COR and CO in writing prior to continuing any manufacturing and/or testing. The Contractor shall provide these documents per Section 5.1 (DRD 1).

#### **4.4 Conferences, Reviews and Surveys Requirements**

The Contractor shall support conferences, reviews, and surveys including corrective action and recurrence controls as specified herein. These reviews and surveys shall be conducted at Contractor facilities and will be chaired by the NASA-JSC COR. The Contractor shall also prepare and make available to the attendees all documentation necessary to accomplish the intended objectives.

##### **4.4.1 Reviews**

The Contractor shall support the following reviews with the necessary documentation and personnel. All reviews shall be conducted by NASA-JSC at the Contractor facility unless other arrangements are approved by the COR.

- **Phase I, Baseline Review:** A Baseline Review shall precede the start of manufacture of any lot of devices. This review shall include the “Control Documentation” referenced in Section 5.1 (DRD 1). The drawings, specifications, parts lists, and/or procedures established as a baseline (control) set by the Phase I review shall become a part of this SOW. NASA-JSC will direct the Contractor with the authorization to proceed with the manufacture of the components for production following this review and the closeout of open review items.
- **Phase II, Production Review:** A Production Review shall precede the start of manufacture of production lot of initiators and/or detonators. This review shall include any changes to the “Control Documentation” referenced in Section 5.1 (DRD

1). This review shall assure adequate evaluation and control of all proposed changes to the baseline and/or last production lot, including the potential effect of the changes upon the qualification status of the devices. The drawings, specifications, parts lists, tooling, and/or procedures established or any changes during the Phase II review shall become a part of this SOW. The review shall also verify that the personnel are currently certified for Pyrotechnic loading, welding, X-ray inspection and N-ray inspection.

*Note: If there is less than 1 year between the acceptance of the last lot and the start of manufacture of the next lot and the last lot and new lot will be identical in all respects, including manufacturing, acceptance, and quality assurance, a production review for the lot may not be required at the discretion of the COR.*

- **Phase III, Lot Acceptance/Certification Review:** A Lot Acceptance Review shall be conducted after the initiators and/or detonators have been manufactured and presented to NASA-JSC for acceptance. This review shall assure that the intent of the Phase I Baseline Review and the Phase II Production Review have been successfully satisfied. The Contractor shall provide the “Lot Acceptance Data Package” in Section 5.2 (DRD 2) and any other applicable documentation requested during the review. The Government shall perform a detailed critical review of all manufacturing, inspection, and acceptance data and records. Acceptance/Certification will be based on successful completion of the review.

#### 4.4.2 Surveys

The Contractor shall support the following surveys with the necessary documentation and personnel. All surveys shall be conducted by NASA-JSC at the Contractor facility unless other arrangements are approved by the COR. For surveys that contain recommended actions, the surveyed organization shall respond within thirty (30) days after receipt of the survey report and submit a status report every thirty (30) days thereafter until all actions are closed, per DRD 3.

- **Safety & Mission Assurance (S&MA) Survey:** The purpose of this survey is to review all safety, reliability and quality assurance documentation. NASA-JSC reserves the right to conduct surveys at any time during the contract. The Contractor shall provide data verifying internal conformance to the S&MA requirements upon request. The Contractor shall establish a schedule of Sub-Contractor SR&QA surveys and shall maintain records of results of surveys which shall be made available to NASA-JSC upon request.
- **Contract & Task Requirements Survey:** The purpose of the survey is to review all contract or task requirements. NASA-JSC reserves the right to conduct one (1) survey at any time during the contract and/or once every twelve (12) months for contracts that exceed twelve (12) months in duration. Thirty (30) days prior to the survey, the CO or COR shall notify the Contractor in writing of the intended survey. This survey will not exceed one (1) week (five (5) working days) duration.

- **Incremental Survey:** The purpose of this survey is to review only a specific area such as a process control problem or any other unsatisfactory detail that could be of such magnitude that the resulting hardware would be unacceptable to NASA-JSC. NASA-JSC reserves the right to conduct an incremental survey at any time to support the resolution of a specific problem. One (1) week (five (5) working days) prior to the incremental survey, the CO or COR shall notify the Contractor either by telecon or writing of the intended survey.

## **4.5 Quality Assurance Requirements**

### **4.5.1 Processes and Products**

The Contractor (and Sub-Contractor designated “critical processes”) shall document changes affecting processes, production equipment, tools and programs. The Contractor shall notify the NASA-JSC COR of proposed changes in process definition and shall obtain approval from the customer prior to implementing the change. The Contractor shall make the documents available for review upon request.

#### 4.5.2 Record Retention

The Contractor and Sub-Contractors shall maintain verifiable objective evidence of all manufacturing, inspections performed, tests performed, results obtained, and dispositions of non-conforming articles. These records shall be identifiable to associated articles, by heat and lot number of materials, unit or lot serialization, and/or other traceable documentation. The Contractor and Sub-Contractors shall make the records available to NASA-JSC upon request. The records shall be retained in a safe, accessible location for a period of ten (10) years after date of delivery as defined in the contract.

*Note: For use in human space flight, the materials, manufacturing, and workmanship of highest quality are essential to astronaut safety. If the Contractor or Sub-Contractors are able to provide the desired item with a higher quality than that of the items specified or proposed, the Contractor and Sub-Contractor are requested to bring this fact to the immediate attention of the customer.*

#### 4.5.3 Procurement Controls

The Contractor shall control and maintain procurement documents to ensure compliance with applicable quality and technical requirements. The procurement documents shall be submitted per Section 5.2 (DRD 2). The Contractor procurements shall include the following:

- **Certificate of Compliance (C of C):** The Sub-Contractor shall provide a certification with each shipment to attest the parts, assemblies, subassemblies, or detail parts conform to the Order Requirements. The certification should include order number, product name, part number, name and address of manufacturing or processing location, product identifiers (e.g. lot, heat, batch, date codes, serial numbers), quantity, unit of measurement and be signed and dated by an official of the company.
- **Raw Material Test Reports:** For raw materials, the Sub-Contractor shall provide chemical analysis reports, physical test reports and/or performance test reports. The test report should list the specification, including revision numbers or letters, to which the material was tested and/or inspected and the identification of the material lot to which it applies.
- **Age Controlled and Limited-Life Products:** For articles and materials having definite characteristics of quality degradation or drift with age, the Sub-Contractor shall provide records that indicate the date and test time or cycle at which useful life was initiated, the life or cycles used, and the date, test time, or cycle at which the useful life is expended.

#### 4.5.4 Raw Material Controls

The Contractor or Sub-Contractors shall inspect and/or test raw materials (e.g. chemical and/or physical analysis and/or performance testing) to determine conformance to applicable drawings and specifications. The reports of actual test results shall identify the particular materials tested. When material specification requires quantitative limits for chemical, mechanical, or physical properties, the test report shall contain the actual test and/or inspection values obtained. When material is altered, the Contractor shall submit all pre and post conversion chemical and/or physical test reports. Raw material shall be segregated and controlled to prevent the use of materials that do not conform to specifications or while awaiting completion and receipt of satisfactory test results.

#### 4.5.5 Traceability

The Contractor shall develop a system to ensure identification of all materials/products, whether separately produced discrete items or material produced in batches, are traceable to the original source/manufacturer and are verifiable. This system shall be maintained throughout the life of this contract, including material/product receipt, all stages of production, delivery, and installation for twenty (20) years past the delivery date.

#### 4.5.6 Problem Reporting and Corrective Action

The Contractor shall provide and maintain a closed loop reporting system involving the Contractor and Sub-Contractors for the reporting of all problems (failures, unsatisfactory conditions, and material review records). The Contractor shall insure that problem reporting, analysis, resolution, and status comply with JSC 28035 and be accomplished in accordance with DRD 4.

- **Immediate Notification:** The Contractor shall report all problems that occur during or subsequent to acceptance to the NASA-JSC COR within 24 hours of occurrence or detection. The Contractor shall also report any problems that occur prior to acceptance testing that will, or have the potential to, adversely affect safety, contribute to the delay of a scheduled event, or result in a design change, to the NASA-JSC COR within 24 hours of occurrence or detection.
- **Problem Analysis:** The supplier shall perform an analysis of each problem reported to determine the cause of the problem and to implement adequate measures to prevent its occurrence. A Contractor shall submit a formal analysis report to the NASA-JSC COR within 14 days of reporting the problem.
- **Problem Resolution:** The Contractor shall document efforts to determine the cause of a problem and the corrective action and implementation in a formal report. The Contractor shall submit the formal report to the NASA-JSC COR and must be approved prior to removal from the Problem Status List.
- **Problem Status List:** The Contractor shall maintain a list of all open problems. The status list shall be submitted per Section 4.10 (DRD 6).

#### **4.5.7 Non-Conforming Material**

The Contractor shall report non-conformances to the customer involving any Contractor and Sub-Contractor materials and/or parts as soon as detected and determined not to be re-workable and may be salvageable. The Contractor shall notify the NASA-JSC COR within three (3) working days after the non-conformance is discovered.

*Note: The customer grants no authority to disposition product or process non-conformances to the Contractor or its Sub-Contractors. Repair is not allowed under this clause.*

#### **4.5.8 Non-Conformance Repot (NCR)**

The Contractor shall provide the NASA-JSC COR with a copy of the NCR or comparable report (e.g. Material Review Report (MRR), Discrepancy Report (DR), etc.) for review. The COR shall coordinate with the cognizant NASA-JSC Quality Assurance personnel to review the NCR. The NASA-JSC COR shall provide written notification to the Contractor of NCR approval/rejection. All NCR dispositions shall be approved by the NASA-JSC COR prior to use (except scrap and return to vendor) per DRD 5.

*Note: A contingent review of the NCR can be obtained by telephone from the COR; however, use of the material prior to formal approval is at the Contractor's risk.*

#### **4.6 Government Source Inspection**

The Contractor shall comply with the following Government Source Inspection requirements.

##### **4.6.1 Mandatory Inspection Points (MIPs)**

The Contractor shall incorporate Mandatory Inspection Points (MIPs), identified by the NASA-JSC Quality Engineer (QE), into Contractor applicable manufacturing and test procedures developed as part of this contract.

##### **4.6.2 Witnessing of Testing**

The Contractor shall allow a NASA-JSC representative or NASA-JSC designated witnesses to observe any and all testing.

##### **4.6.3 Notification of Testing**

The Contractor shall notify the NASA-JSC COR (or approved representative) a minimum of forty-eight (48) hours in advance of intended subject testing of the test procedures specified herein. The Contractor shall only perform MIP identified testing in the presence of a NASA-JSC approved representative.

#### **4.6.4 Discrepancy**

The Contractor shall immediately contact NASA-JSC COR in the event that a Government representative witnessing testing has identified any incompatibilities between hardware, processes, and/or technical requirements. The Contractor shall immediately initiate a non-conformance action in the case of discrepant hardware or request contract change if required to eliminate any incompatibilities.

#### **4.7 Verification of Test Records & Manufacturing**

Verification of Quality Control Surveillance shall be indicated by the identifying mark (inspection stamp) of the Government/Contractor Quality Control personnel or authorized delegates on the test data sheets as applicable.

#### **4.8 Safety Requirements**

The Contractor shall adhere to the following safety requirements.

- The Contractor shall comply with all federal, state, and local codes and standards applicable to work performed on this contract.
- In the event of serious injury or illness to personnel or damage to equipment and/or property arising from this contract, the Contractor shall notify the Occupational Safety & Institutional Assurance Division, code NS, at 281-483-7500 (Safety Action Hotline), within twenty-four (24) hours of such occurrence. The Contractor shall conduct an investigation into the causes of the accident and implement corrective measures. However, NASA-JSC reserves the right to conduct investigation into such occurrence as necessary. The Contractor shall provide support as required.
- The Contractor shall perform hazard analyses in sufficient depth to identify and correct conditions which could cause serious personal injury or illness, or significant damage to property limited to manufacturing and processing of toxic materials, test facilities systems, test set-ups, test operations, radio frequency hazards, auto-ignition temperatures, pollution, and environmental contamination.

#### **4.9 Control of Inspection, Measuring, and Test Equipment**

The Contractor shall have a documented calibration system that meets the requirements of American National Standard Institute (ANSI)/National Conference of Standards Laboratories (NCSL) Z540.3-2006 (R2013) Requirements for Calibration of Measuring and Test Equipment.



#### **4.10 Program Status Report**

The Contractor shall provide the COR with a monthly status update in accordance with DRD 6. The Contractor shall include the following at a minimum:

- Schedule
- Production Status
- Open Action Items
- Upcoming Reviews
- Problem Status List

*Note: The monthly status report may be replaced by weekly meetings with Contractor provided minutes at the discretion of the COR.*

### **5.0 DOCUMENTATION REQUIREMENTS**

The Contractor shall furnish documentation as set forth in the Data Requirements List (DRL) and the Data Requirements Documents (DRDs).

#### **5.1 Control Documentation**

The Contractor shall provide NASA-JSC the record set of all documents pertaining to the materials, manufacturing, quality control, and acceptance of the devices per DRD 1.

- The record set shall include all drawings (including tooling), quality control procedures, process specifications, manufacturing specifications, key personnel, and any other documentation used in the production and testing of the devices.
- The record set shall also include all changes to the documents and a corresponding change-effectivity matrix.
- The record set shall establish the baseline for the production of the devices and shall be maintained throughout the duration of the contract.
- Each document, within the record set, shall be approved by the NASA-JSC COR prior to implementation.
- The Contractor shall provide a document list that includes the document number, title, revision number, and revision date of all documents in the record set. This document list shall encompass both Contractor and Sub-Contractor documents and revisions thereto.
- The Contractor shall provide NASA-JSC with one copy of each approved document within the record set.

## 5.2 Lot Acceptance Data Package

The Contractor shall provide NASA-JSC a lot acceptance data package per DRD 2. The Contractor shall make the lot acceptance data package available for review by the NASA-JSC representatives during at the lot acceptance review (Phase III). The Contractor shall ship the lot acceptance data package to NASA-JSC after acceptance of the lot. The lot acceptance data package shall include the following at a minimum:

- Certified acceptance reports including the date of manufacture of the devices in the lot and the lot number of the explosive material utilized.
- Certified list of all detail piece parts by their applicable drawing numbers, including revision letters. Receiving inspection records with numbers of units acceptance, to provide lot production and/or rejection traceability.
- Phase Review Meeting Minutes
- Documented final inspection records, including the original set of X-Ray and N-Ray films of each part in the lot, with copies of the N-Ray and X-Ray certification prepared by the performing vendor.
- Copies of destructive lot acceptance data, including copies of the pressure/time traces and any other applicable data.
- Material coupon tensile test results, requirements and proof pressure tests results with performing vendor certification.
- Statement certifying the formula of each explosive mix contained in each unit is the same used for qualification. Caloric test data for the powder in the lot/batch.
- The recorded weights including pre-load and post-load weights of each increment of the explosive mix.
- Copies of all nondestructive lot acceptance data including leak test information, bridgewire resistance readings and any other applicable data.
- Copies of all failure and corrective action records (e.g. NCRs, MRRs, DRs, waivers, deviations, etc.).
- A copy of the explosive classification. The Contractor of each pyrotechnic device shall be responsible for obtaining the explosive classification for that device from the U.S. Department of Transportation.
- A copy of the Competent Authority Approval. The Contractor of each pyrotechnic device shall be responsible for obtaining the Competent Authority Approval Letter, for that device from the U.S. Dept. of Transportation.
- A copy of the Material Safety Data Sheet. The Contractor for each pyrotechnic device shall be responsible for obtaining the Material Safety Data Sheet on the current OSHA form designated for that device being presented for certification.
- Lot Certificate. Prepared by NASA-JSC COR before shipment.
- Receiving inspection records of piece parts with material and process certification data sheets.
- Vendor certification records pertaining to material traceability from raw stock.
- A copy of each completed operational/manufacturing record (traveler) used in production of subject lot with inspection stamps therein denoting acceptance.

### **5.3 Post-Production Modifications Acceptance Data Package**

The Contractor shall provide NASA-JSC a post-production modifications acceptance data package per DRD 7. The Contractor shall ship the post-production modifications acceptance data package to NASA-JSC when the post-production modifications are shipped. The post-production modifications acceptance data package shall include the following at a minimum:

- Certified acceptance reports including the date of manufacture of the devices in the lot and the lot number of the explosive material utilized.
- Certified list of all detail piece parts by their applicable drawing numbers, including revision letters. Receiving inspection records with numbers of units acceptance, to provide lot production and/or rejection traceability.
- Documented final inspection records, including helium leak check test results.
- Copies of all nondestructive lot acceptance data including leak test information, bridgewire resistance readings and any other applicable data.
- Copies of all failure and corrective action records (e.g. NCRs, MRRs, DRs, waivers, deviations, etc.).
- A copy of the explosive classification. The supplier of each pyrotechnic device shall be responsible for obtaining the explosive classification for that device from the U.S. Department of Transportation.
- A copy of the Competent Authority Approval. The supplier of each pyrotechnic device shall be responsible for obtaining the Competent Authority Approval Letter, for that device from the U.S. Dept. of Transportation.
- A copy of the Material Safety Data Sheet. The supplier for each pyrotechnic device shall be responsible for obtaining the Material Safety Data Sheet on the current OSHA form designated for that device being presented for certification.
- Receiving inspection records of piece parts with material and process certification data sheets.
- Vendor certification records pertaining to material traceability from raw stock.
- A copy of each completed operational/manufacturing record (traveler) used in production of subject lot with inspection stamps therein denoting acceptance.

## **6.0 SHIPPING**

### **6.1 Shipping Containers**

NASA-JSC shall provide the Contractor with shipping containers, P/N SDZ38121697-301, for the shipment of the initiators to JSC.

NASA-JSC shall not provide the Contractor with shipping containers. Packaging of detonators for shipment will be performed per Vendor documentation. Vendor documentation will be approved by NASA prior to implementation.

## **6.2 Shipping Instructions**

### **6.2.1 Devices & Loaded Pyrotechnics**

The Contractor shall ship all deliverable devices to:

NASA/Lyndon B. Johnson Space Center  
Transportation Officer, Bldg. 420  
2101 NASA Parkway  
Houston, TX. 77058  
Attn: Jason Wolinsky / EP6 / B352 / x49656

### **6.2.2 Documentation Deliverables**

The Contractor shall ship all deliverable documentation, including N-Rays Films, to:

NASA/Lyndon B. Johnson Space Center  
Transportation Officer, Bldg. 420  
2101 NASA Parkway  
Houston, TX. 77058  
Attn: Todd Hinkel / EP3 / B15-R126 / x30251

### **6.2.3 GSE**

The Contractor shall return GSE to:

NASA/Lyndon B. Johnson Space Center  
Transportation Officer, Bldg. 420  
2101 NASA Parkway  
Houston, TX. 77 058  
Attn: Refer to JF290 Block 1 / B352

*Note: For return shipments to JSC, include a copy of the JF290. A copy of the JF290 will be shipped with the hardware and can also be requested from the COR.*